

Development of Marketing Potential and Cost Management for Stingless Bee Honey Products to Enhance the Income of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province

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Abstract

This research, titled "Development of Marketing Potential and Cost Management for Stingless Bee Honey Products to Enhance the Income of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province," aims to (1) develop the marketing potential and cost management of stingless bee honey products, and (2) build knowledge and skills in product management among community members in order to add value and generate sustainable income for the community enterprise group.

The study employed an action research design through workshop activities on cost calculation, income–expense recording, product pricing, and the development of online marketing channels via a Facebook Page and a LINE Official Account for members of the 789 Ban Kha Community Enterprise Group in Ratchaburi Province. The products were also trialled for sale in collaboration with Inum Café in order to gather consumer feedback.

The findings revealed that members of the community enterprise gained greater knowledge and understanding of cost management and marketing, enabling them to apply this knowledge effectively in product development, pricing, and expanding distribution channels. In addition, the stingless bee honey products attracted greater consumer interest, which allowed the community enterprise to generate income and strengthen the local economy in a sustainable manner.

Keywords: stingless bee honey, community enterprise, marketing, cost management, community development

Introduction

At present, stingless bee honey products have received increasing attention from consumers because they are natural products with nutritional value and contain antioxidants beneficial to the body. They also have properties that help boost the immune system and have become popular among health-conscious consumers (Somchai et al., 2021; Department of Agricultural Extension, 2022). Furthermore, stingless bee keeping is an occupation that can continuously generate income for farmers and local communities. As a result, many communities have begun to develop stingless bee farming and processed products in order to add economic value and increase community income (Office of Agricultural Economics, 2022).

However, although stingless bee honey products are gaining popularity, many community enterprise groups still face challenges in cost management, product pricing, and marketing—particularly in promoting their products through online channels. This has resulted in their products not being widely known and being unable to compete effectively in the market (Office of Community Enterprise Promotion, 2021). In addition, most community members still lack knowledge of cost calculation, income–expense planning, and value creation for their products, which affects long-term income and economic stability (Nattawut et al., 2020).

Today, online marketing is considered an important factor in expanding access to consumers and can effectively help small entrepreneurs and community enterprises broaden their distribution channels. In particular, social media platforms such as Facebook and LINE Official Account provide convenient and rapid access to consumers (Kotler & Keller, 2019; Thanaphat, 2022). Therefore, developing knowledge of online marketing alongside cost management is an important approach to enhancing the competitive potential of community products.

The 789 Ban Kha Community Enterprise Group in Ratchaburi Province is one of the communities that produce and sell stingless bee honey as a community product, but it still has limitations regarding cost management and online marketing. These limitations affect the expansion of distribution channels and the group's ability to generate income for its members. Accordingly, this research aims to study approaches to developing the marketing potential and cost management of stingless bee honey products through the transfer of knowledge on cost management, product pricing, and online distribution channels, in order to add value to the product, generate income, and sustainably strengthen the economic foundation of the community enterprise group.

Research Objectives

1. To develop the marketing potential and cost management of stingless bee honey products of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province, in order to add value and tangibly generate income for community members.
2. To build knowledge and skills in the management of stingless bee honey products for the community, leading to occupational development, self-reliance, and the long-term sustainability of the community enterprise group.

Literature Review

1. Concepts of Cost Management and Community Business Administration

1.1 Definition of Cost Management

Cost management refers to the process of planning, controlling, and analyzing the expenses incurred in producing goods and services so that the business can operate efficiently and generate reasonable profits (Samuelson & Nordhaus, 2010). Effective cost management enables entrepreneurs to calculate production costs, set product prices, and plan production appropriately, while also reducing waste from unnecessary use of resources.

In the context of community enterprises, cost management is a crucial factor for generating income and ensuring the community's economic stability, as most entrepreneurs still lack systematic income–expense recording and cost calculation systems (Burapha University, 2017).

1.2 The Importance of Cost Management for Community Enterprises

Research by Nattawut et al. (2020) found that community enterprises with systematic cost management can plan production and generate profits more effectively than groups without clear cost management. Furthermore, accurate cost calculation enables product prices to be set in line with costs and market conditions, which affects the competitiveness of community products.

Therefore, transferring knowledge on cost management to community members is highly important in helping entrepreneurs operate their businesses more efficiently and generate income sustainably.

2. Concepts of Marketing and Communication for Community Products

2.1 The Marketing Mix Concept (4Ps)

Marketing is a key process in creating value and expanding economic opportunities for community products. The marketing mix, or 4Ps, consists of Product, Price, Place, and Promotion (Kotler & Keller, 2016). These concepts can be applied to the development of stingless bee honey products in order to make them more attractive and build consumer confidence.

2.2 Online Marketing and Digital Media

Today, online media play an important role in consumer purchasing decisions, particularly platforms such as Facebook and LINE Official Account, which can be used effectively as channels for promotion and sales (Kotler & Keller, 2019).

Research by Thanaphat Suwanchai (2022) found that online marketing increases opportunities to reach consumers and significantly boosts sales for community products. Meanwhile, research by Prapatsorn Chindarat (2023) found that price, ordering convenience, and communication through online media influence consumer purchasing decisions.

The above concepts and research findings reflect that the development of online marketing channels is an important approach to enhancing the competitive potential of community products today.

3. Concepts Regarding Stingless Bee Honey Products

3.1 The Value of Stingless Bee Honey

Stingless bee honey is a natural product with nutritional value and high levels of antioxidants, which are beneficial to physical health (Biluca et al., 2016). Moreover, stingless bee honey has chemical properties that differ from those of regular honey and can be developed into high-value-added health products (Kek et al., 2014).

In Thailand, stingless bee farming has become increasingly popular because it can generate income for farmers and communities, and the product aligns with the current trend toward health-conscious consumption (Department of Agricultural Extension, 2022).

3.2 Related Research on Stingless Bee Honey Products

Most research, both domestic and international, has focused on the nutritional value of stingless bee honey and the development of products derived from it. However, research on cost management and the marketing of stingless bee honey products at the community level remains relatively limited.

Therefore, this study aims to apply marketing and cost management concepts together with the development of online distribution channels to add value and generate income for the 789 Ban Kha Community Enterprise Group in Ratchaburi Province.

4. Concepts of Packaging Development for Community Products

Packaging is an important element that adds value and builds credibility for products. In addition to protecting goods, it serves as a medium for communicating key information to consumers, such as product quantity, ingredients, and properties (Lockamy, 2003).

Research by Burapha University (2017) found that many community products still have limitations in packaging design and incomplete product information disclosure, which affects consumer confidence and market competitiveness.

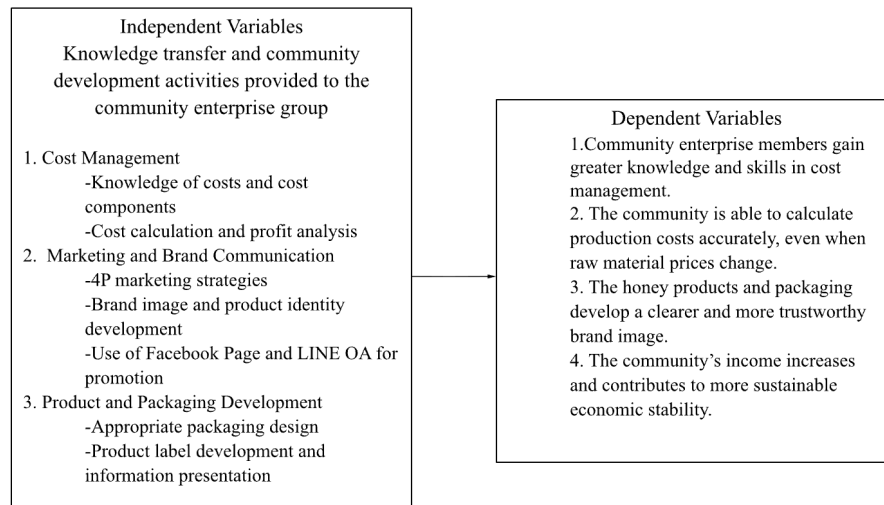
In the case of the 789 Ban Kha Community Enterprise Group in Ratchaburi Province, it was found that, in the initial phase, stingless bee honey products were sold in packaging that did not clearly display product information. Therefore, developing packaging with complete information and an appropriate format is another important approach to adding value and building product credibility.

5. Concepts of Developing the Potential of Community Enterprises

Developing the potential of community enterprises is an important process for strengthening economic resilience and self-reliance within communities, drawing on the transfer of knowledge, skills development, and the creation of marketing opportunities for community members (Office of Community Enterprise Promotion, 2021).

The concept of sustainable community development emphasizes the use of local resources and wisdom to generate income and improve the quality of life of community members through their own efforts. Therefore, developing knowledge of marketing, cost management, and value addition for stingless bee honey products is a key approach to strengthening the economic stability and long-term sustainability of the 789 Ban Kha Community Enterprise Group in Ratchaburi Province.

Conceptual Framework



Research Methodology

1. Type of Research

This study is qualitative research employing a developmental research approach to examine problems, needs, and approaches for developing cost management and online marketing among the target group. Data were collected through in-depth interviews, observation, and actual field testing in order to formulate suitable approaches for practical application.

2. Population and Sample

2.1 Population

The population in this study comprised small entrepreneurs, students, and individuals interested in online product sales and cost management.

2.2 Sample

The sample consisted of 9 villagers in the area, selected through purposive sampling from individuals who were interested in or involved with product sales, cost management, and the use of online media for product promotion. They served as informants and participants in the field testing, providing feedback on the development approaches in this study.

3. Issues Examined

This study focused on the following issues:

1. Problems and needs regarding the cost management of the target group.
2. Approaches for using online media to promote and sell products.
3. Problems, obstacles, and suggestions arising from actual implementation.

4. Approaches for developing cost management and online marketing models appropriate for the target group.

4. Research Instruments

The instruments used for data collection included:

1. A semi-structured interview form for collecting data on the problems, needs, and suggestions of the target group.
2. An observation form for recording behaviors and usage patterns during the activities.
3. A cost, income, and expense recording form for studying the cost management practices of participants.
4. Digital tools, including LINE Official Account (LINE OA) and Facebook Page, for trialling product promotion and customer communication in real situations.

5. Research Procedure

The implementation was divided into 4 phases as follows

Phase 1: Studying Information and Problems

Reviewed relevant research, documents, and information, and collected data through interviews and observation in order to examine the problems, needs, and behaviors of the target group regarding cost management and online product sales.

Phase 2: Developing Approaches and Activities

Used the data obtained from the preliminary study to develop approaches for cost management and the use of online media for product promotion, and designed trial activities appropriate for the target group.

Phase 3: Actual Implementation

Trialled online platforms such as LINE OA and Facebook Page for product promotion and sales in collaboration with a café in order to study usage patterns, problems, and limitations that emerged in practice.

Phase 4: Summarizing and Refining the Approach

Compiled data from interviews, observations, and participant feedback for analysis and the formulation of refined approaches for cost management and online marketing.

6. Data Analysis

The researchers used content analysis to analyze data from interviews, observations, and participant feedback. The data were categorized, key issues were analyzed, and conclusions were drawn in line with the objectives of the research.

Results

Part 1: Developing the Marketing Potential and Cost Management of Stingless Bee Honey Products of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province, in Order to Tangibly Add Value and Generate Income for Community Members

The findings revealed that, in the first redesign of the stingless bee honey product packaging both the bottle and the pouch-with-box format the use of more luxurious color tones made the product more attractive and better met consumer needs in terms of usability. When the products were trialled at a trade fair, all the prepared stock sold out within the event, and most consumers commented that the packaging was beautiful, looked luxurious, and that the pouch format was more convenient to use. However, this package received feedback from designers indicating that it did not clearly resemble a honey product, both in terms of the logo, color tone, and overall design.

Figure1: Comparison of stingless bee honey product packaging before and after development



Before development



First version



Latest version

The packaging was therefore developed into a final version using more natural color tones, with a new bee-themed logo to communicate the product more clearly, and decorations featuring flowers that serve as food for the community's stingless bees. This resulted in a package with a distinctive identity unique to Ban Kha. Designers gave feedback that this package clearly communicated the honey theme, and community members voted in favor of the new packaging, which was adopted as the main package for actual sales. When the brand was introduced into online sales channels, it was found that four bottles of honey could be sold online, generating revenue of 1,440 baht—approximately 12 percent of total sales. This reflects the potential for expanding into online markets and represents a promising starting point for brand development

The images show two pages of handwritten revenue records. The left page lists quarterly revenue for B.E. 2568 (Q1: 21,560, Q2: 12,400, Q3: 19,930, Q4: 36,583) and B.E. 2569 (Q1: 8,300, Q2: 25,570*). The right page provides a detailed breakdown of these quarterly totals into monthly figures, with handwritten calculations for each quarter's total.

(Figure2: Revenue from stingless bee honey sales, B.E. 2568–2569 [2025–2026]. Quarterly revenue: Q1/2568 = 21,560 baht; Q2/2568 = 12,400 baht; Q3/2568 = 19,930 baht; Q4/2568 = 36,583 baht; Q1/2569 = 8,300 baht; Q2/2569* = 25,570 baht. *Note: Q2/2569 data are not yet complete for the full quarter.)



The graph shows that, following the packaging development and participation in two product fairs, community income in Q4 of B.E. 2568 (2025) increased markedly, reflecting that improvements to the product format combined with activity-based marketing promotion helped to boost consumer interest and expand sales opportunities. Furthermore, after launching the latest version of the packaging and introducing the product into the online market in Q2 of B.E. 2569 (2026), revenue showed a renewed upward trend, even though the data for this quarter were not

yet complete. These results reflect that developing marketing potential—in terms of packaging design, expansion of distribution channels, and entry into the online market—plays an important role in tangibly increasing income and adding value for the community.

Regarding sales promotion, it was found that running discount promotions through online channels made products more accessible to consumers, resulting in higher sales volumes and creating opportunities to expand the customer base and increase community income.

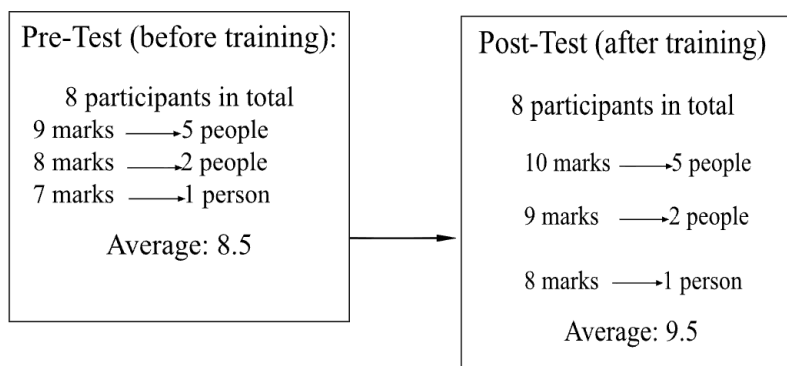
Part 2: Building Knowledge and Skills in the Management of Stingless Bee Honey Products for Community Members in Order to Sustainably Add Value and Generate Income for the Community Enterprise Group



The findings revealed that after the team built knowledge of cost calculation, pricing, and the use of technology via the LINE OA platform among members of the community enterprise group, the team evaluated the outcomes both before and after the training using interviews and tests (pre-test and post-test) to measure the level of knowledge and understanding of community members.

The evaluation results showed that, after the training, community members had greater knowledge and understanding of cost and profit calculation, with the average post-training test score increasing by 1 point compared with the pre-training score. In addition, post-training interviews revealed that members were able to calculate costs and set selling prices in a systematic, step-by-step manner, enabling them to clearly identify appropriate price levels and profit margins. As a result, the community could confidently set new prices when needed, which gave them the confidence to run discount promotions during the initial phase of online sales to make products more accessible to customers.

Illustration of pre- and post-training test results:



Furthermore, the knowledge gained was not limited to stingless bee honey products. Members were also able to apply the principles of cost calculation to other community businesses already in operation, such as pineapple and chili paste products. In addition, members were able to learn and use LINE OA technology as a primary channel for promotion, customer communication, and online product sales, thereby expanding market opportunities and sustainably generating income for the community in the future.

Discussion

The research findings showed that the development of the marketing potential of the stingless bee honey product of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province, through packaging improvement and the creation of a brand identity could clearly add value to and elevate the product's image. In the first phase of packaging development, the products were designed in both bottle and pouch-with-box formats to appear more luxurious and user-friendly, attracting consumer interest and selling out at the trade fair. This reflects that packaging design has an impact on consumer interest and purchasing decisions. However, although the packaging received positive feedback for its attractiveness, designers also commented that its format, colors, and elements did not clearly communicate that it was a honey product. This indicates that effective packaging design must not only be visually distinctive but also appropriately reflect the identity and category of the product.

Based on these findings, the research team developed the final version of the packaging, adopting natural color tones, designing a bee-shaped logo, and using floral elements that serve as food for the local stingless bees to decorate the packaging—communicating the product identity more clearly and better reflecting the spirit of the Ban Kha community. The result was that this new packaging gained acceptance from both designers and community members, with the community voting to adopt it as the main packaging for actual sales. This reflects that developing products with attention to community identity and the participation of stakeholders can enhance product value and acceptance. It is also consistent with marketing concepts that view packaging as a key element in building brand awareness, credibility, and perceived value, particularly for community products competing in broader markets.

Regarding online marketing, introducing products into online channels such as Facebook Page and LINE Official Account opened opportunities to reach new consumer groups and facilitated communication and ordering. Although the early online sales trial resulted in only 4 bottles sold, generating 1,440 baht—about 12 percent of total sales—this is still a significant starting point that reflects the potential for expanding into digital markets. In addition, running discount promotions through online channels increased product accessibility for consumers, contributing to the creation of new customer bases and an increase in community income during the early stages. This reflects the role of digital technology in supporting the growth of community enterprises in the present era.

In terms of building knowledge and skills in product management, the pre- and post-training evaluations revealed that community members had greater knowledge and understanding of cost calculation, pricing, and profit management. The average score increased by 1 point after the training. Interviews also revealed that members were able to calculate costs and set selling prices more systematically, enabling them to understand appropriate price levels and plan profits more clearly. As a result, the community gained confidence in setting product prices and was able to formulate pricing strategies—such as running discount promotions in the initial stage of online sales—more rationally. This indicates that the workshop training effectively built skills that could be applied in practice to community business management.

Moreover, the knowledge gained from the training was not limited to the production and sale of stingless bee honey. Community members could also apply cost calculation principles to other community products, such as pineapple and chili paste, and use LINE OA as a tool for promotion, customer communication, and as a sales channel. This reflects the potential of knowledge development to be extended into income generation and the long-term strengthening of the community economy.

Nevertheless, some limitations were also encountered during project implementation, including differences in age range and basic knowledge among community members, which affected the learning process and the transfer of knowledge—requiring varied communication approaches and more time to establish mutual understanding. In addition, although the development of packaging and product image helped to make the product more attractive, price remained a significant factor influencing consumer purchasing decisions. This reflects that product development should go hand in hand with cost management, appropriate pricing, and marketing strategy formulation, so that community products can compete effectively and grow sustainably in the future.

Summary

Based on the project's implementation, it can be concluded that the development of the stingless bee honey product of the 789 Ban Kha Community Enterprise Group, Ratchaburi Province—covering marketing, packaging design, and the building of knowledge on cost management—was able to tangibly add value and create income-generating opportunities for the community.

In terms of developing the marketing potential and cost management, the redesign of both the bottle and the pouch-style packaging made the product appear more modern, of higher value, and better suited to consumer needs, resulting in stronger sales performance. In the first sales trial, the stock sold out within the event, and consumers gave positive feedback that the product appeared luxurious, was easy to use, and more suitable for urban customers. Furthermore, introducing the product into the online market opened up new sales opportunities, with online sales accounting for approximately 12 percent of total sales—a promising starting point for market expansion. In addition, running discount promotions also helped consumers access the product more easily and influenced purchasing decisions.

In terms of building knowledge and skills in product management, after the training, community members had a clearly greater understanding of cost calculation, pricing, and the use of LINE OA technology. The post-training assessment showed a higher average score than before, and members were able to apply this knowledge systematically in pricing and profit planning. The knowledge of cost calculation could also be extended to other community products such as pineapple and chili paste. At the same time, learning to use LINE OA gave the community a more effective channel for promotion, communication with customers, and online sales—an important foundation for generating income and sustainably developing community potential in the future.

Limitations

1. Limitations in Production

The community is still unable to control the quantity of stingless bee honey production to meet market demand consistently, as the production process depends on natural factors, seasons, and the number of bee hives available. As a result, when market demand is high, products may be insufficient or unable to be delivered on time, which could affect consumer confidence and opportunities to expand the market in the future.

2. Limitations Regarding Product Standards

The community's products have not yet received certification from relevant agencies, such as the Food and Drug Administration (FDA) or the Community Product Standard (CPS / มาตรฐาน.). This may result in consumers, particularly those purchasing through online channels, lacking confidence in the quality, safety, and credibility of the products, which could affect purchasing decisions and the expansion of the customer base.

3. Limitations in Communication and Marketing

The community still has limitations in communication and product promotion skills—particularly in highlighting product strengths, creating engaging media or content, and using online media for effective marketing. In addition, the coordination and communication between project implementers and community members sometimes required time to develop mutual understanding, which caused some parts of the work to fall behind the research schedule.

Recommendations

1. Production

Given the limitations regarding inconsistent production quantity, the community should plan production to align more closely with market demand—for example, by increasing the number of stingless bee hives, providing appropriate hive maintenance, and systematically planning seasonal harvesting. In addition, stock management and clear sales cycles should be implemented, including the possible adoption of a pre-order system, in order to reduce the risk of product shortages and build long-term consumer confidence.

2. Product Standards

The community should pursue certification from relevant agencies, such as the Food and Drug Administration (FDA) or the Community Product Standard (CPS), in order to increase credibility and build consumer confidence, particularly in online channels. In addition, attention should be given to developing packaging and label information that is complete, accurate, and standard-compliant, which would help elevate the professional image of the product.

3. Communication and Marketing

The community should continue to develop skills in communication and digital marketing—such as creating attractive promotional content, taking product photos, presenting product strengths, using social media effectively, and responding to customer enquiries appropriately and promptly. In addition, marketing communication should be clearly planned to help expand customer reach, build brand awareness, and promote a positive product image in the long term.

4. Market Expansion

Once the community has effectively developed its production capacity, product standards, and online marketing, it should plan to expand into new customer markets, both domestic and international. This may start with participating in trade fairs, building business networks, or introducing products into online platforms that reach overseas markets. Preparing in terms of product standards, production, and brand identity will be key factors in increasing competitive opportunities and ensuring the community's sustainable growth in the future.

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